



NORTH SYDNEY BOYS HIGH SCHOOL

2016 HSC ASSESSMENT TASK 3 (TRIAL HSC)

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- For Section I, shade the correct box on the sheet provided
- For Section II, write in the booklet provided
- Each new question is to be started on a **new page**.
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Ireland
- ☐ Mr Lin
- ☐ Dr Jomaa

Student Number

(To be used by the exam markers only.)

Question No	1-10	11	12	13	14	15	16	Total	Total
Mark	$\frac{\quad}{10}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{100}$	$\frac{\quad}{100}$

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10 located at the end of this section.

- 1 If $z = 2 - 5i$ find z^{-1} expressed with a real denominator
- (A) $\frac{1}{24}(2 + 5i)$
- (B) $\frac{1}{29}(2 + 5i)$
- (C) $\frac{1}{29}(2 - 5i)$
- (D) $\frac{1}{24}(-2 + 5i)$
- 2 If the line $y = mx + b$ is a tangent to the hyperbola $xy = c^2$, which of the following is true?
- (A) $b^2 = -4mc^2$
- (B) $b^2 = 4mc^2$
- (C) $b = 4mc$
- (D) $c^2 = 4mb$
- 3 For a certain function $y = f(x)$, the function $f(|x|)$ is represented by:
- (A) A reflection of $y = f(x)$ in the y axis
- (B) A reflection of $y = f(x)$ in the x axis
- (C) A reflection of $y = f(x)$ in the x axis for $y \geq 0$
- (D) A reflection of $y = f(x)$ in the y axis for $x \geq 0$

4 A hyperbola has equation $x^2 - 4y^2 = 4$. The distance between its directrices is:

(A) $\sqrt{5}$

(B) $\frac{4\sqrt{5}}{5}$

(C) $2\sqrt{5}$

(D) $\frac{8\sqrt{5}}{5}$

5 If $e^x + e^y = 1$, $\frac{dy}{dx} =$

(A) $-e^{x-y}$

(B) e^{y-x}

(C) e^{x-y}

(D) $-e^{y-x}$

6 The polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ has real coefficients, and

$P(2i) = P(2 + i) = 0$ What is $a + b + c + d$?

(A) 0

(B) 1

(C) 4

(D) 9

- 7 The solutions to the equation $x^4 - 10x^2 + 5 = 0$ are

$$x = \tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5}, \tan \frac{4\pi}{5}$$

What is the value of: $\tan^2 \frac{\pi}{5} + \tan^2 \frac{2\pi}{5} + \tan^2 \frac{3\pi}{5} + \tan^2 \frac{4\pi}{5}$?

- (A) 20
- (B) 5
- (C) -20
- (D) 10

- 8 Which expression is equal to $\int \frac{1}{1 - \sin x} dx$

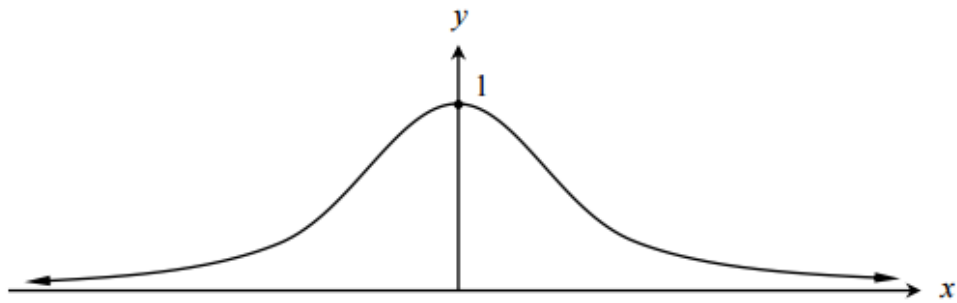
- (A) $\tan x - \sec x + c$
- (B) $\tan x + \sec x + c$
- (C) $\log_e(1 - \sin x) + c$
- (D) $\frac{\log_e(1 - \sin x)}{-\cos x} + c$

- 9 Let ω be the complex root of unity such that $\omega^n = 1$, $\omega \neq 1$

Find the value of $\sum_{k=0}^n (w^k + \frac{1}{w^k})$

- (A) 0
- (B) 1
- (C) 2
- (D) 3

10 Which of the following equations best describe this curve?



(A) $y = e^{x^2}$

(B) $y = \frac{1}{(x+1)^2}$

(C) $y = \frac{1}{x^2 + 1}$

(D) $y = e^{-x^2-x}$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section.

Answer each question on a NEW page. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW page in your writing booklet.

(a) Find $\int \frac{(\ln x)^2}{x} dx$. 2

(b) Find $\int \sin^3 x \cos^2 x dx$ 2

(c) Find $\int \frac{dx}{\sqrt{3-2x-x^2}}$ 2

(d) By using the substitution of $x = 2 \tan \theta$, evaluate $\int_0^2 \frac{dx}{(4+x^2)^{\frac{3}{2}}}$ 2

(e) (i) Express $\frac{7x}{(x^2+3)(x+2)}$ in the form of $\frac{Ax+B}{x^2+3} + \frac{C}{x+2}$ 2

(ii) Hence evaluate $\int_0^3 \frac{7x}{(x^2+3)(x+2)} dx$ 2

(f) The area bounded by the curve $y = 2 \cos x$ for $0 \leq x \leq 2\pi$ and the line $y = 2$ is rotated about the y axis. Find the volume of the solid formed using the method of cylindrical shells. 3

Question 12 (15 marks) Start a NEW page in your writing booklet.

- (a) For the hyperbola $\frac{y^2}{25} - \frac{x^2}{16} = 1$, find
- (i) the eccentricity 1
 - (ii) the coordinates of the foci S and S' and the equations of its directrices 2
 - (iii) Sketch the hyperbola showing all the above features. 1
- (b) Explain why the equation $\frac{x^2}{\lambda - 23} + \frac{y^2}{5 - \lambda} = 1$ cannot represent the equation of an ellipse. 1
- (c) Let $f(x) = x^2(x - \frac{3}{2})$ be a function on the domain $-1 \leq x \leq 2$.
- (i) Draw a neat sketch of $y = f(x)$, labelling all intersections with the coordinate axes and turning points. You are not required to test the nature of the turning points. 2
 - (ii) Sketch $y = \frac{1}{f(x)}$. 2
 - (iii) Sketch $y^2 = f(x)$ 2
- (d) Find the coordinates of the stationary points for the curve $x^2 - 4xy = y^2 - 20$ 4

Question 13 (15 marks) Start a NEW page in your writing booklet.

(a) Shade the region defined by the intersection of $|\arg z| \leq \frac{\pi}{3}$, $z + \bar{z} \leq 4$ and $|z| \geq 2$ 3

(b) If $z = \sqrt{3} + i$

(i) Find the exact value of $|z|$ and $\arg z$ 2

(ii) By using De Moivre's theorem write $\frac{1}{z^5}$ in the form of $x + iy$ 2

(c) A polynomial function is $P(x) = x^5 + x^4 + 13x^3 + 13x^2 - 48x - 48$.

Factorise $P(x)$ over the field of:

(i) real numbers, 2

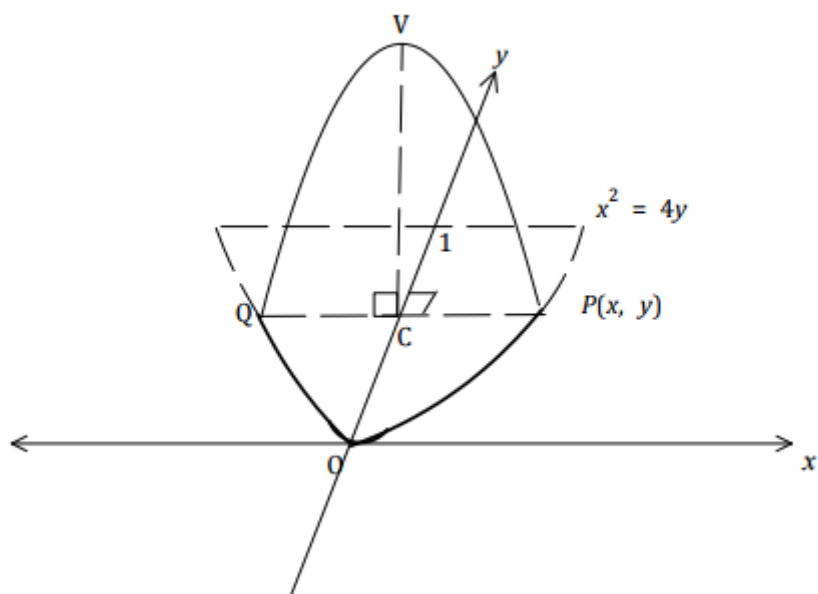
(ii) complex numbers. 1

(d) Find the complex square roots of $7 + 6i\sqrt{2}$, giving your answer in the form of $a + ib$, where a and b are real. 2

(e) R is a positive number and z_1, z_2 are complex numbers. Show that the points on the Argand diagram which represent respectively the numbers $z_1, z_2, \frac{z_1 - iRz_2}{1 - iR}$ form the vertices of a right angled triangle. 3

Question 14 (15 marks) Start a NEW page in your writing booklet.

(a)



The base of the solid is the region bounded by the parabola $x^2 = 4y$ and the line $y = 1$.

Cross sections perpendicular to this base and the y axis are parabolic segments with their vertices V directly above the y axis. The diagram shows a typical segment PVQ . All the segments have the property that the vertical height VC is three times the base length PQ .

Let $P(x, y)$ where $x \geq 0$ be a point on the parabola $x^2 = 4y$

(i) Show that the area of the segment PVQ is $8x^2$. 2

(ii) Hence, find the volume of the solid. 2

Question 14 continues on Page 11

Question 14 (continued)

(b) let $z = \cos \theta + i \sin \theta$ be any complex number of modulus 1.

(i) Show that $\frac{z^2 - 1}{z} = 2i \sin \theta$ 2

(ii) Hence, prove that 2

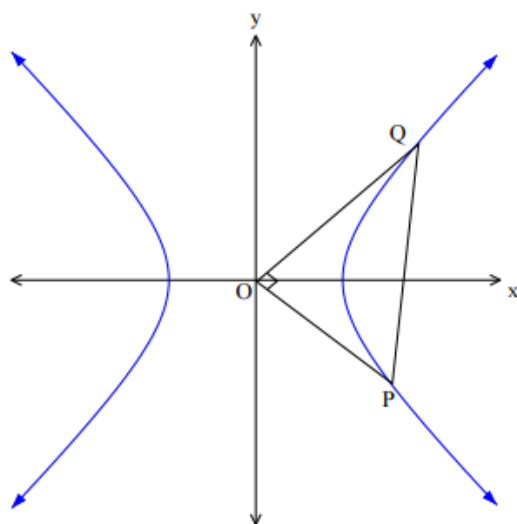
$$z + z^3 + z^5 + z^7 + z^9 = \frac{\sin 10\theta + i(1 - \cos 10\theta)}{2 \sin \theta}$$

(iii) Hence write down a simplified expression for 1

$$\cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta + \cos 9\theta$$

(c) The diagram shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$.

The points $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \alpha, b \tan \alpha)$ lie on the hyperbola and the chord PQ subtends a right angle at the origin.



(i) Show that $\sin \theta \sin \alpha = -\frac{a^2}{b^2}$. 3

(ii) Hence show that the gradient of the curve at $P(a \sec \theta, b \tan \theta)$ is 3

$$\frac{dy}{dx} = -\frac{b^3}{a^3} \sin \alpha$$

Question 15 (15 marks) Start a NEW page in your writing booklet.

(a) Let $I_n = \int_0^1 x^n \ln(1+x) dx, n = 0, 1, 2, \dots$

(i) Show that $\int \ln(1+x) dx = (1+x) \ln(1+x) - x + c$ 1

(ii) Show that $(n+1)I_n = 2 \ln 2 - \frac{1}{n+1} - nI_{n-1}, n = 1, 2, \dots$ 2

(iii) Evaluate $3I_2$ and $4I_3$ 2

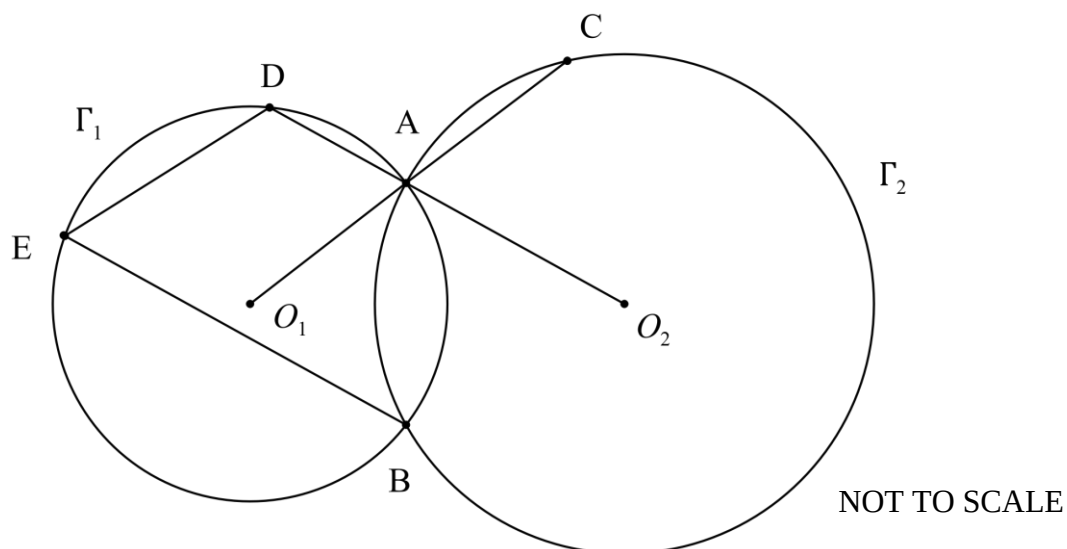
(iv) Show that $(n+1)I_n = \begin{cases} \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1}, & \text{for } n \text{ is odd} \\ 2 \ln 2 - \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n+1} \right), & \text{for } n \text{ is even} \end{cases}$ 2

(b) Given $S_n = \sum_{k=1}^n k^2$. Prove by mathematical induction that:

$$nS_n - \sum_{r=1}^{n-1} S_r = \sum_{r=1}^n r^3 \text{ for } n > 1$$

Question 15 continued on Page 13

(c)



In the figure, the two circles Γ_1 and Γ_2 have centres O_1 and O_2 respectively. They intersect at the points A and B . The extensions of O_1A meets Γ_2 at C and the extension of O_2A meets Γ_1 at D .

Given that $BE \parallel O_2A$ and $ED \parallel O_1A$. Let $\angle BED = \alpha$

(i) Copy or trace the diagram into your answer booklet.

(ii) Prove that O_1DCO_2 is a cyclic quadrilateral 2

(iii) Hence, prove that $DC \perp CO_2$ 2

Question 16 (15 marks) Start a NEW page in your writing booklet.

- (a) (i) Given that $\frac{a+b}{2} \geq \sqrt{ab}$ 1

Prove that $\frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$

- (ii) Using part (i) and the fact that $\frac{a+b+c}{3} = \frac{1}{4} \left(a+b+c + \frac{a+b+c}{3} \right)$, 2
 prove that $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$

- (b) (i) Prove that for all positive values of x , $e^x > 1 + x$ 3

- (ii) If $x_1, x_2, \dots, x_n$ are positive numbers with $S_n = x_1 + x_2 + \dots + x_n$ and 2
 $R_n = (1+x_1)(1+x_2) \dots (1+x_n)$, show that $e^{S_n} > R_n > 1 + S_n$

- (iii) Let $b_k = \frac{k^2 + k + 1}{k^2 + k}$ and $P_n = b_1 \times b_2 \times \dots \times b_n$ 3

Show that $P_n < e$ for any positive whole numbers n

- (c) Given a polynomial $P(x) = x^3 - ax^2 + bx - a$ where a and b are positive real numbers. Let a be the smallest positive real number such that all the roots of the polynomial are positive and real.

- (i) Explain why all three roots of the polynomial are positive real numbers. 1

- (ii) Using the result from part (a) (ii), find the value of a and b 3

End of Examination

Multiple Choice.

1. B 2. A 3. D 4. D 5. A 6. D 7. A
8. B 9. C 10. C

Question 11

a) $I = \int (\ln x)^2 \cdot \frac{1}{x} dx$

By using the reverse chain rule

$$I = \frac{(\ln x)^3}{3} + C$$

b) $I = \int \sin^3 x \cos^2 x dx$

$$= \int \sin x (1 - \cos^2 x) \cos^2 x dx$$

$$= \int \sin x \cos^2 x - \sin x \cos^4 x dx$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

c) $\int \frac{dx}{\sqrt{3 - (2x + x^2)}} = \int \frac{dx}{\sqrt{4 - (x+1)^2}}$

$$= \sin^{-1}\left(\frac{x+1}{2}\right) + C$$

d) let $x = 2 \tan \theta$

when $x = 0$ $\theta = 0$

when $x = 2$ $\theta = \frac{\pi}{4}$

$$dx = 2 \sec^2 \theta d\theta$$

$$\therefore I = \int_0^{\frac{\pi}{4}} \frac{2 \sec^2 \theta}{(4 + 4 \tan^2 \theta)^{\frac{3}{2}}} d\theta$$

$$= \frac{1}{4} \int_0^{\frac{\pi}{4}} \cos \theta d\theta$$

$$= \frac{1}{4} \left[\sin \theta \right]_0^{\frac{\pi}{4}}$$

$$= \frac{\sqrt{2}}{8}$$

e) $\frac{7x}{(x^2+3)(x+2)} \equiv \frac{Ax+B}{x^2+3} + \frac{C}{x+2}$

$$(Ax+B)(x+2) + C(x^2+3) \equiv 7x$$

when $x = -2$

$$C = -2$$

$$A+C=0 \Rightarrow A=2$$

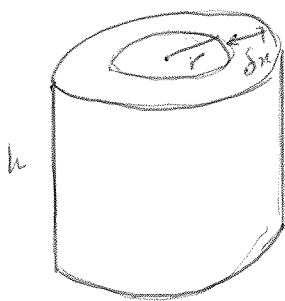
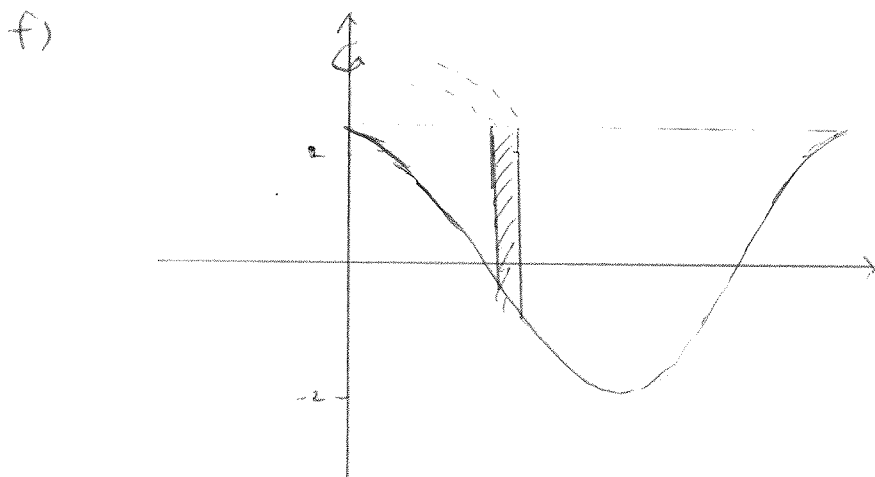
when $x = 0$

$$2B+3C=0$$

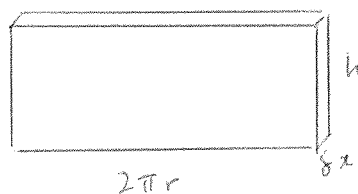
$$B=3$$

$$\frac{7x}{(x^2+3)(x+2)} \equiv \frac{2x+3}{x^2+3} - \frac{2}{x+2}$$

$$\begin{aligned}
 \text{(i)} \quad \int_0^3 \frac{7x}{(x^2+3)(x+2)} dx &= \int_0^3 \frac{2x+3}{x^2+3} - \frac{2}{x+2} dx \\
 &= \int_0^3 \frac{2x}{x^2+3} + \frac{3}{x^2+3} - \frac{2}{x+2} dx \\
 &= \left[\log(x^2+3) + \frac{3}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} - 2 \log|x+2| \right]_0^3 \\
 &= \log 12 + \frac{\pi}{\sqrt{3}} - \log 25 - \log 3 + \log 4 \\
 &= \log \left(\frac{16}{25} \right) + \frac{\pi}{\sqrt{3}}
 \end{aligned}$$



This volume is equivalent to



$$\begin{aligned}
 \delta V &= 2\pi r h \delta x \\
 &= 2\pi x \cdot (2 - 2\cos x) \delta x \\
 &= 4\pi x (1 - \cos x) \delta x
 \end{aligned}$$

$$\begin{aligned}
 \therefore V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{2\pi} 4\pi x (1 - \cos x) \\
 &= 4\pi \int_0^{2\pi} x (1 - \cos x) dx
 \end{aligned}$$

$$\begin{aligned}
 &= 4\pi \left[x(x - \sin x) \right]_0^{2\pi} \\
 &\quad - \int_0^{2\pi} x - \sin x dx \\
 &= 4\pi \left[4\pi^2 - \left[\frac{x^2}{2} + \cos x \right]_0^{2\pi} \right] \\
 &= 4\pi [4\pi^2 - 2\pi^2] \\
 &= 8\pi^3 u^3
 \end{aligned}$$

Question 12

a) (i) $a^2 = b^2(e^2 - 1)$

$$16 = 25(e^2 - 1)$$

$$\frac{16}{25} = e^2 - 1$$

$$e^2 = \frac{41}{25}$$

$$e = \frac{\sqrt{41}}{5}$$

(ii) Foci

$$(0, \pm 5 \times \frac{\sqrt{41}}{5})$$

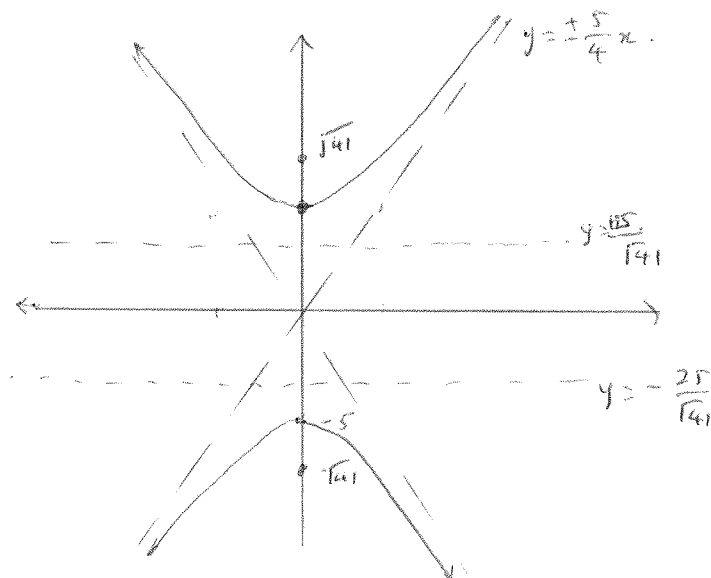
$$= (0, \pm \sqrt{41})$$

Directrices

$$y = \pm \frac{5}{\frac{\sqrt{41}}{5}}$$

$$= \pm \frac{25}{\sqrt{41}}$$

(iii) asymptotes are $y = \pm \frac{5}{4}x$. vertices are $(0, 5)$, $(0, -5)$



b) For an ellipse $\lambda - 23 > 0$ $5 - \lambda > 0$
 $\therefore \lambda > 23$ $5 > \lambda$

There are no values of λ that will satisfy this condition

$$\therefore \frac{x^2}{\lambda - 23} + \frac{y^2}{5 - \lambda} = 1$$

cannot represent the equation of an ellipse.

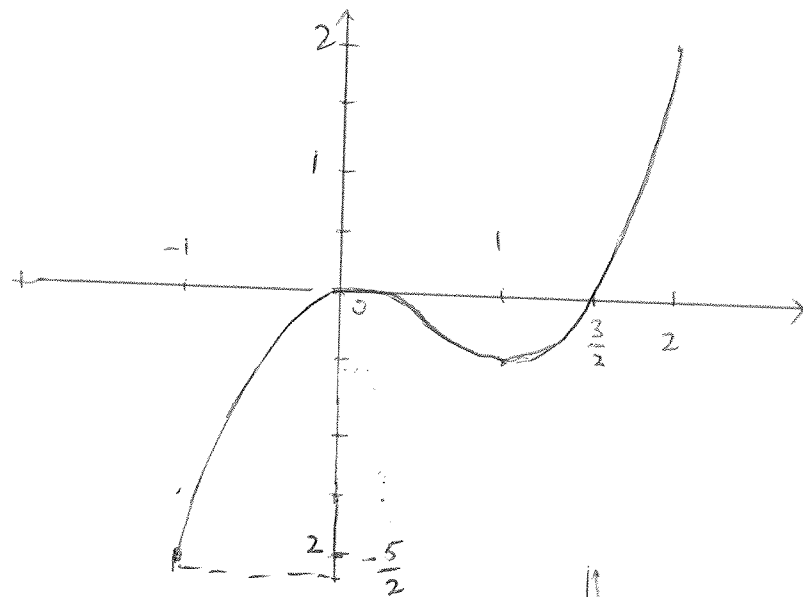
c) (i) $f(x) = x^2(x - \frac{3}{2}) \quad -1 \leq x \leq 2$

$$= x^3 - \frac{3x^2}{2}$$

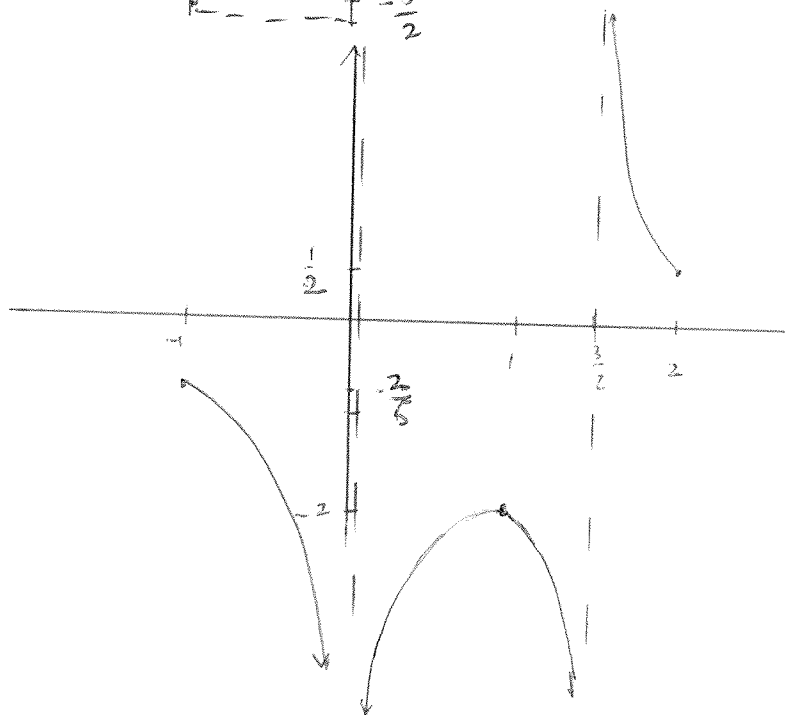
$$f'(x) = 3x^2 - 3x$$

$$= 3x(x-1)$$

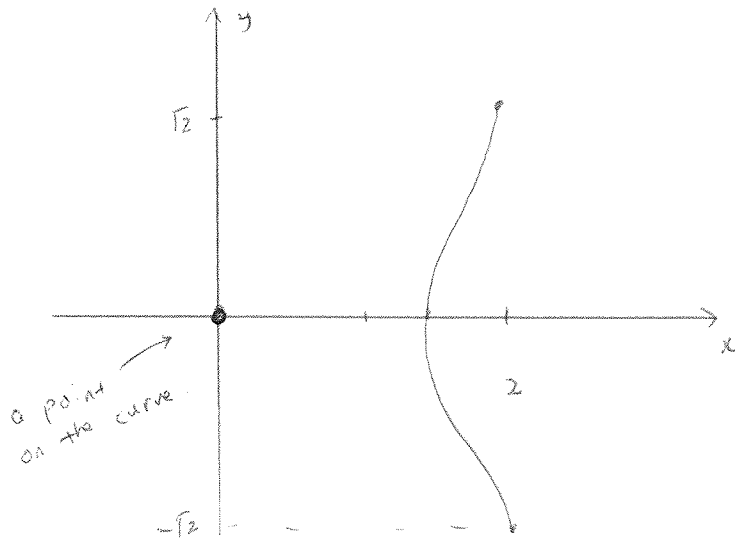
∴ stationary points are at $(0,0)$ and $(1, -\frac{1}{2})$



(ii)



iii)



$$d) \quad 2x - \left(4x \frac{dy}{dx} + 4y\right) = 2y \frac{dy}{dx}$$

$$2x - 4x \frac{dy}{dx} - 4y = 2y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{2x - 4y}{2y + 4x}$$

$$= \frac{x - 2y}{y + 2x}$$

stationary points when $x - 2y = 0$

$$\text{i.e. } x = 2y$$

$$\therefore 4y^2 - 8y^2 = y^2 - 20$$

$$5y^2 = 20$$

$$y^2 = 4$$

$$y = \pm 2$$

$$\therefore \text{when } y = 2, x = 4$$

$$y = -2, x = -4$$

$\therefore$ stationary points are

$$(4, 2), (-4, -2)$$

Question 13

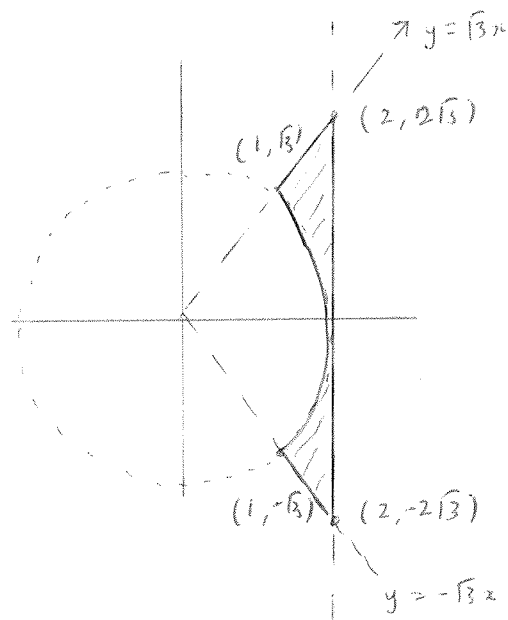
a) $z + \frac{1}{z} \leq 4$

$\therefore 2x \leq 4$

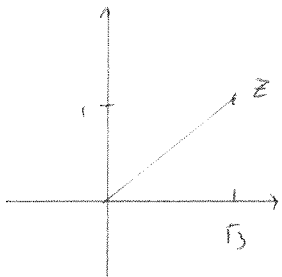
$x \leq 2$

$|\arg z| \leq \frac{\pi}{3}$

$\therefore -\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{3}$



b)



(i) $|z| = \sqrt{3+1}$
 $= 2$

$\arg z = \frac{\pi}{6}$

(ii) $z = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$

$\therefore z^{-5} = \left[2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right]^{-5}$

By De Moivre's Thm.

$z^{-5} = \frac{1}{32} \left(\cos \left(-\frac{5\pi}{6} \right) + i \sin \left(-\frac{5\pi}{6} \right) \right)$

$= \frac{1}{32} \left(\cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} \right)$

$= -\frac{\sqrt{3}}{64} - \frac{1}{64}i$

(c) i) $P(x) = x^4(x+1) + 13x^2(x+1) - 48(x+1)$

$= (x+1)(x^4 + 13x^2 - 48)$

$= (x+1)(x^2-3)(x^2+16)$

$= (x+1)(x-\sqrt{3})(x+\sqrt{3})(x^2+16)$

ii) $P(x) = (x+1)(x-\sqrt{3})(x+\sqrt{3})(x-4i)(x+4i)$

$$d) \quad z^2 = 7 + 6\sqrt{2}i$$

$$\text{let } z = a + ib$$

$$\therefore (a+ib)^2 = a^2 - b^2 + 2abi$$

$$\therefore a^2 - b^2 = 7$$

$$2ab = 6\sqrt{2}$$

$$ab = 3\sqrt{2}$$

$$a^2 + b^2 = 11$$

$$\therefore 2a^2 = 18$$

$$a^2 = 9$$

$$a = \pm 3 \quad \therefore b = \pm \sqrt{2}$$

$$\therefore z = \pm (3 + \sqrt{2}i)$$

$$e) \quad \arg \left(z_1 - \left(\frac{z_1 - R iz_2}{1 - Ri} \right) \right)$$

$$= \arg \left(\frac{Ri(z_2 - z_1)}{1 - Ri} \right)$$

and

$$\arg \left(z_2 - \left(\frac{z_1 - R iz_2}{1 - Ri} \right) \right)$$

$$= \arg \left(\frac{z_2 - z_1}{1 - Ri} \right)$$

where $z_3 = \frac{z_1 - R iz_2}{1 - Ri}$

$$z_3 = \frac{z_1 - R iz_2}{1 - Ri}$$

$$\text{Now } \arg(z_1 - z_3) = \arg(z_2 - z_3)$$

$$= \arg \left(\frac{Ri(z_2 - z_1)}{1 - Ri} \right) = \arg \left(\frac{z_2 - z_1}{1 - Ri} \right)$$

$$= \arg \left(\frac{Ri(z_2 - z_1)}{z_2 - z_1} \right)$$

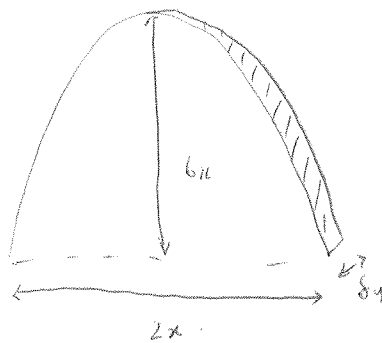
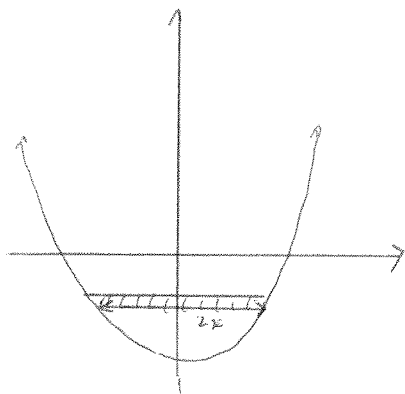
$$= \arg(Ri)$$

$$= \frac{\pi}{2} \quad \text{as } R > 0.$$

$\therefore z_1, z_2, z_3$ form a right angle triangle.

Question 14

(i)



using Simpson's rule

$$\begin{aligned} A(x) &= \frac{2x}{3} (0 + 4 \times 6x + 0) \\ &= \frac{24x^2}{3} \\ &= 8x^2 \end{aligned}$$

(ii)

$$\begin{aligned} \delta V &= 8x^2 \delta y \\ &= 8(4y) \delta y \end{aligned}$$

$$\begin{aligned} V &= \lim_{\delta y \rightarrow 0} \sum_{y=0}^1 32y \delta y \\ &= 32 \int_0^1 y \, dy \\ &= 32 \left[\frac{y^2}{2} \right]_0^1 \\ &= 16 \text{ units}^3 \end{aligned}$$

14 b) (i) $\frac{z^2-1}{z} = z - \frac{1}{z}$ let $z = \cos\theta + i\sin\theta$ as modulus is 1

$$= (\cos\theta + i\sin\theta) - (\cos\theta + i\sin\theta)^{-1}$$

$$= \cos\theta + i\sin\theta - (\cos(-\theta) + i\sin(-\theta)) \quad \text{By De Moivre's Thm}$$

$$= \cos\theta + i\sin\theta - (\cos\theta - i\sin\theta)$$

$$= 2i\sin\theta$$

(ii) $z + z^3 + z^5 + z^7 + z^9$ is a geometric progression

where the common ratio $r = z^2$, first term is $a = z$ and $n = 5$

$$\therefore S_5 = \frac{z((z^2)^5 - 1)}{z^2 - 1}$$

$$= \frac{z}{z^2 - 1} (z^{10} - 1)$$

$$= \frac{1}{2i\sin\theta} (\cos 10\theta + i\sin 10\theta - 1) \quad \text{From part (i)}$$

$$= \frac{i\cos 10\theta - \sin 10\theta - i}{-2\sin\theta}$$

$$= \frac{\sin 10\theta + i(1 - \cos 10\theta)}{2\sin\theta}$$

(iii) Hence $\cos\theta + \cos 3\theta + \cos 5\theta + \cos 7\theta + \cos 9\theta$ is the real part of the expression found in part (ii)

$$\therefore \cos\theta + \cos 3\theta + \cos 5\theta + \cos 7\theta + \cos 9\theta = \frac{\sin 10\theta}{2\sin\theta}$$

c) (i) Gradient of chord OQ

$$m_{OQ} = \frac{b \tan \alpha}{a \sec \alpha} = \frac{b \sin \alpha}{\cos \alpha} \times \frac{\cos \alpha}{a}$$
$$= \frac{b \sin \alpha}{a}$$

Similarly gradient of chord OP

$$m_{OP} = \frac{b \sin \theta}{a}$$

Since $OQ \perp OP$

$$m_{OQ} \times m_{OP} = -1$$

$$\therefore \frac{b \sin \alpha}{a} \times \frac{b \sin \theta}{a} = -1$$

$$\frac{b^2 \sin \alpha \sin \theta}{a^2} = -1$$

$$\sin \alpha \sin \theta = -\frac{a^2}{b^2}$$

(ii) Differentiating implicitly

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

Substituting the point P.

$$= \frac{b^2 a \sec \alpha}{a^2 b \tan \alpha}$$

$$= \frac{b \sec \alpha}{a \tan \alpha}$$

$$= \frac{b}{a \sin \alpha}$$

Since $\sin \theta = -\frac{a^2}{b^2 \sin \alpha}$

$$\therefore \frac{1}{\sin \theta} = -\frac{b^2 \sin \alpha}{a^2}$$

$$\therefore \frac{dy}{dx} = \frac{b}{a} \times \frac{-b^2 \sin \alpha}{a^2}$$
$$= -\frac{b^3}{a^3} \sin \alpha.$$

Question 15

$$a) \quad I_n = \int_0^1 x^n \ln(1+x) dx, \quad n=0, 1, 2.$$

$$i) \quad I_0 = \int_0^1 1^0 \ln(1+x) dx$$

$$= x \ln(1+x) - \int \frac{x}{1+x} dx$$

$$= x \ln(1+x) - \int 1 - \frac{1}{1+x} dx$$

$$= x \ln(1+x) - x + \ln(1+x) + C.$$

$$= (x+1) \ln(1+x) - x + C.$$

$$ii) \quad I_n = \left[x^n (1+x) \ln(1+x) - x \right]_0^1 - n \int_0^1 x^{n-1} \left[(1+x) \ln(1+x) - x \right] dx$$

$$= 2 \ln 2 - 1 - n \int_0^1 (x^{n-1} + x^n) \ln(1+x) - x^n dx$$

$$= (2 \ln 2 - 1) - n \int_0^1 x^{n-1} \ln(1+x) + x^n \ln(1+x) - x^n dx$$

$$= 2 \ln 2 - 1 - n (I_{n-1} + I_n) + n \int_0^1 x^n dx$$

$$= 2 \ln 2 - 1 - n I_{n-1} - n I_n + n \left[\frac{x^{n+1}}{n+1} \right]_0^1$$

$$= 2 \ln 2 - 1 - n I_{n-1} - n I_n + \frac{n}{n+1}$$

$$I_n + n I_n = 2 \ln 2 - 1 + \frac{n}{n+1} - n I_{n-1}$$

$$\therefore I_n (n+1) = 2 \ln 2 + \frac{-n-1+n}{n+1} - n I_{n-1}$$

$$I_n = 2 \ln 2 - \frac{1}{n+1} - n I_{n-1}$$

iii)

$$I_0 = \int_0^1 \ln(1+x) dx$$

$$= \left[(1+x) \ln(1+x) - x \right]_0^1$$

$$= (2 \ln 2 - 1)$$

$$2I_1 = 2 \ln 2 - \frac{1}{2} - I_0$$

$$= 2 \ln 2 - \frac{1}{2} - (2 \ln 2 - 1)$$

$$= 1 - \frac{1}{2}$$

$$3I_2 = 2 \ln 2 - \frac{1}{3} - 2I_1$$

$$= 2 \ln 2 - \frac{1}{3} - (1 - \frac{1}{2})$$

$$= 2 \ln 2 - 1 + \frac{1}{2} - \frac{1}{3}$$

$$4I_3 = 2 \ln 2 - \frac{1}{4} - 3I_2$$

$$= 2 \ln 2 - \frac{1}{4} - (2 \ln 2 - 1 + \frac{1}{2} - \frac{1}{3})$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4}$$

$$iv) (n+1) I_n = 2 \ln 2 - \frac{1}{n+1} - n I_{n-1}$$

$$= 2 \ln 2 - \frac{1}{n+1} - \left[2 \ln 2 - \frac{1}{n} - (n-1) I_{n-2} \right]$$

$$= -\frac{1}{n+1} + \frac{1}{n} + (n-1) I_{n-2}$$

when n is odd the last term will be $2I_1$, i.e. $n=3$.

$$\text{i.e. } 2I_1 = 1 - \frac{1}{2}$$

$\therefore$ no log term.

when n is even the last term will be I_0

where $I_0 = (2\ln 2 - 1)$

$$(n+1)I_n = \begin{cases} 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1}, & \text{for } n \text{ is odd} \\ 2\ln 2 - \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n+1}\right) & \text{for } n \text{ is even} \end{cases}$$

b) Given $S_n = \sum_{k=1}^n k^2$

RTP $nS_n - \sum_{r=1}^{n-1} S_r = \sum_{r=1}^n r^3$ for $n > 1$

Base case when $n=2$.

$$\text{LHS} = 2S_2 - \sum_{r=1}^1 S_r$$

$$= 2(1^2 + 2^2) - 1^2$$

$$= 9$$

$$\text{RHS} = \sum_{r=1}^2 r^3$$

$$= 1^3 + 2^3$$

$$= 1 + 8$$

$$= 9$$

$\therefore \text{LHS} = \text{RHS} \quad \therefore \text{true for } n=2.$

Assume true for $n=k$ i.e. $k \in \mathbb{Z}$

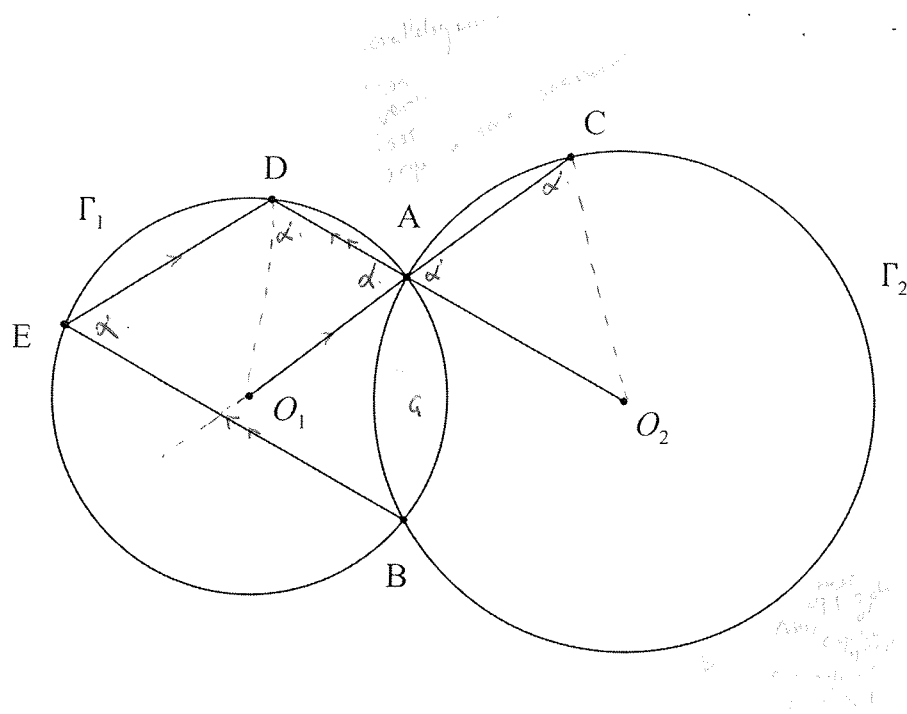
$$kS_k - \sum_{r=1}^{k-1} S_r = \sum_{r=1}^k r^3$$

Required to prove true for $n=k+1$

i.e. $(k+1)S_{k+1} - \sum_{r=1}^k S_r = \sum_{r=1}^{k+1} r^3$

$$\text{LHS} = (k+1)S_{k+1} - \sum_{r=1}^k S_r$$

$$= (k+1)[S_k + (k+1)^2] - \sum_{r=1}^{k-1} S_r - S_k.$$



(i) Extending O_1A to line EB , let the point of intersection be F

∴ $ADEF$ is a parallelogram (2 pairs of parallel sides)

∴ $\angle BAO_1 = \alpha$ (opposite angles of a parallelogram are equal)

$\angle CAO_2 = \alpha$ (vertically opposite angles are equal).

∴ $\triangle AO_1D$ and $\triangle AO_2C$ are isosceles (as $O_1D = O_1A$ and $O_2A = O_2C$)

$\angle O_1DA = \alpha$ and $\angle ACO_2 = \alpha$ (base angles of isosceles triangle are equal)

∴ $\angle O_1DA = \angle ACO_2$

∴ O_1DCO_2 is a cyclic quadrilateral (equal angles subtended by equal chords in the same segment)

Join O_1O_2 and AB . Intersect at G .

(ii) ∴ $\triangle O_1AG \equiv \triangle O_1BG$ (SSS)

$\triangle O_1AG \equiv \triangle O_1BG$ (SAS)

∴ $O_1O_2 \perp AB$ ($\angle O_1GA = \angle BGO_1$ and AGB is a straight angle)

$\angle DAG = 180 - \alpha$ (opposite angles of cyclic quadrilateral $ADEB$)

∴ $\angle O_1AG = 180 - 2\alpha$

∴ $\angle AO_1G = 2\alpha - 90^\circ$ (angle sum of triangle)

∴ $\angle DO_1A = 180 - 2\alpha$ (angle sum of triangle)

then $\angle DO_1G = 90^\circ \Rightarrow \angle DCO_2 = 90^\circ$ (opposite angles of cyc. quad are supplementary)

Question 16

$$a) i) \quad \frac{a+b}{2} + \frac{c+d}{2} \geq \sqrt{ab} + \sqrt{cd}$$

applying the AM-GM inequality again

$$\geq 2 \sqrt{\sqrt{ab} \cdot \sqrt{cd}}$$

$$= 2 (abcd)^{\frac{1}{4}}$$

$$\therefore \frac{a+b+c+d}{4} \geq \sqrt[4]{abcd}$$

$$(ii) \quad \text{Given } \frac{a+b+c}{3} = \frac{1}{4} \left(a+b+c + \frac{a+b+c}{3} \right)$$

here we can use part (i)

$$\frac{a+b+c}{3} \geq \sqrt[4]{abc \left(\frac{a+b+c}{3} \right)}$$

$$\frac{(a+b+c)}{3} \geq (abc)^{\frac{1}{4}} \cdot \left(\frac{a+b+c}{3} \right)^{\frac{1}{4}}$$

$$\left(\frac{a+b+c}{3} \right)^{\frac{3}{4}} \geq (abc)^{\frac{1}{4}}$$

$\therefore$ taking powers of $\frac{4}{3}$ on both sides

$$\left(\frac{a+b+c}{3} \right) \geq (abc)^{\frac{1}{3}}$$

C (i) Any valid explanation will suffice. (Conjugate root theorem)

(i) Let the roots be p, q, r .

$$\therefore pqr = a \quad p+q+r = a$$

applying the AM-GM.

$$\frac{p+q+r}{3} \geq \sqrt[3]{pqr}$$

$$\frac{a}{3} \geq \sqrt[3]{a}$$

$$\therefore \frac{a^2}{27} \geq 1$$

$$a \geq 3\sqrt{3}$$

$\therefore$ smallest positive value of a is $3\sqrt{3}$.

$\therefore b = 9$ (From product of roots 2 at a time)